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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

MA2004-1 – VECTOR CALCULUS & TRANSFORM TECHNIQUES

(For AC, EC, EE, VL)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. If $\phi = x^2 + y - z - 1$, then $|\nabla \phi|$ at $(1, 0, 0)$ is
A) $\sqrt{5}$ B) $\sqrt{2}$
C) $\sqrt{6}$ D) 1
2. The unit normal vector to the curve $x^2 + y^2 - 2z + 3 = 0$ at $(1, 2, -1)$ is
A) $\frac{2\hat{i}+4\hat{j}+2\hat{k}}{\sqrt{24}}$ B) $\frac{2\hat{i}+4\hat{j}}{\sqrt{20}}$
C) $\frac{2\hat{i}+2\hat{j}}{\sqrt{20}}$ D) $\frac{2\hat{i}+4\hat{j}-2\hat{k}}{\sqrt{24}}$
3. The value of $\nabla \times (xy\hat{i} + yz\hat{j} + zx\hat{k})$ is
A) $-y\hat{i} - z\hat{j} - x\hat{k}$ B) $-y\hat{i} + z\hat{j} - x\hat{k}$
C) $-y + z - x$ D) $-y - z - x$
4. $\vec{F}$ is said to be solenoidal if
A) $\nabla \cdot \vec{F} = 0$ B)
C) $\nabla \times \vec{F} = \vec{0}$ D) $\nabla \times \vec{F} \neq \vec{0}$
D) $\nabla \cdot (\nabla \times \vec{F}) = 0$
5. Let ϕ be a scalar potential of the conservative vector field $\vec{F}$. Then the work done in moving a particle along a curve from point A to B is
A) $\phi(B) - \phi(A)$ B) 0
C) $\phi(B) + \phi(A)$ D) $\phi(A) - \phi(B)$
6. Work done in moving a particle along the straight line $y = x$ from A (0,0) to B (1,1) under the force $\vec{F} = x^2\hat{i} + xy\hat{j}$ is
A) 0 B) 2/3
C) 1/3 D) 7/5
7. Using the Green's theorem, the value of the line integral $\int_C y^2 dx + x^2 dy$, where C is the boundary of the square $-1 \leq x \leq 1, -1 \leq y \leq 1$ is
A) 4 B) $2(x + y)$
C) 0 D) $\frac{4}{3}$
8. Stoke's theorem relates
A) line and surface integral B) surface and volume integral
C) line and volume integral D) line integral, surface integral and volume integral
9. If $w = \frac{az+b}{cz+d}$ is a bilinear transformation, then
A) $ab - dc \neq 0$ B) $ad - bc = 0$
C) $ad - bc \neq 0$ D) $ab - dc = 0$

10. The residue of $f(z) = \frac{z^3+8}{(z-1)(z+2)}$ at $z = 1$ is
 A) 0 B) 3
 C) 9 D) 2
11. If $f(z) = x + ay + i(bx + cy)$ is analytic, then the values of a, b, c are
 A) $a = 1$ and $c = -b$ B) $b = 1$ and $a = -c$
 C) $a = b = c = 1$ D) $c = 1$ and $a = -b$
12. The simple poles of $f(z) = \frac{z^2-4}{z^2+5z+4}$ are
 A) 1, 4 B) 1, -4
 C) -1, -4 D) -1, 4
13. If $f(z) = u + iv$ where $z = x + iy$ then $\int_C f(z) dz$ can be expressed as
 A) $\int_C (u dx + v dy) - i \int_C (v dx + u dy)$ B) $\int_C (u dx - v dy) + i \int_C (u dx + v dy)$
 C) $\int_C (u dx - v dy) + i \int_C (v dx + u dy)$ D) $\int_C (u dx + v dy) + i \int_C v dx - u dy$
14. According to Cauchy's integral formula, if $f(z)$ is analytic everywhere in a region R bounded by a closed curve C then for $a \in R$, $\oint_C \frac{f(z)}{z-a} dz$ is equal to
 A) $2\pi i$ B) $f(a)$
 C) $2\pi i f(a)$ D) 0
15. The poles of $f(z) = \frac{1}{(z^2+1)^3}$ are
 A) $\pm 3i$ B) $\pm i$
 C) 1, 3 D) 2, 3
16. The value of $\oint_C e^z dz$ around a unit circle $|z| = 1$ is
 A) 0 B) 1
 C) $1/2$ D) 2
17. The half range Fourier sine series expansion of $f(x)$ in $(0, 1)$ is
 A) $f(x) = \sum_{n=1}^{\infty} b_n \sin n x$ B) $f(x) = \sum_{n=1}^{\infty} b_n \sin n \pi x$
 C) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} b_n \sin n \pi x$ D) $f(x) = \sum_{n=1}^{\infty} b_n \sin 2 \pi x$
18. If $f(x) = x^2$ in $-\pi < x < \pi$, $f(x+2\pi) = f(x)$ then the value of the Fourier coefficient b_n is
 A) 0 B) $\frac{2\pi^2}{3}$
 C) $\frac{\pi^2}{3}$ D) $\frac{2\pi}{3}$
19. The unilateral Z-transform of 6^n is
 A) $\frac{z}{z-6}$ B) $\frac{z}{z+6}$
 C) $\frac{z-1}{z-6}$ D) $\frac{z}{z-n}$
20. If $u_n = \begin{cases} 2^n & \text{if } n \geq 0 \\ 0 & \text{if } n < 0 \end{cases}$ and $Z(u_n) = U(z)$ then $\lim_{z \rightarrow \infty} U(z)$ equals
 A) 1 B) 0
 C) 2 D) -1

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at the time $t = 1$ in the direction of vector $\hat{i} - 3\hat{j} + 2\hat{k}$.	6	L*2	1	1
	b) Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction of the vector $2\hat{i} - \hat{j} - 2\hat{k}$.	5	L2	1	1
	c) Using divergence theorem of Gauss evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S be the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.	5	L2	2	1

2. a) If $\phi = 2x^3y^2z^4$ and $\vec{F} = \nabla\phi$, find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$. Evaluate these at (1, -1, 1). 6 L1 1 1
- b) Find the work done in moving a particle once around the circle in the xy- plane if the circle has centre at the origin and radius 3, given that the force field is $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y)\hat{k}$. 5 L2 2 1
- c) Show that the vector field $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is a conservative force field. Find its scalar potential. 5 L2 2 1
3. a) State and prove Green's theorem in a plane. 6 L2 2 1
- b) Prove that the cylindrical co-ordinate system is an orthogonal curvilinear coordinate system. 5 L2 1 1
- c) Determine the constants a, b and c so that the function $\vec{F} = (axy + z^3)\hat{i} + (bx^2 + z)\hat{j} + (bxz^2 + cy)\hat{k}$ is irrotational. 5 L1 1 1

Unit – II

4. a) If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then prove that at any point of D, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. 6 L2 3 1
- b) Find the bilinear transformation which maps $z = \infty, i, 0$ onto $w = -1, -i, 1$. 5 L1 3 2
- c) State Cauchy's integral formula. Using Cauchy's integral formula, evaluate $\int_C \frac{e^z}{(z+1)^2} dz$, if C is the circle $|z - 1| = 3$. 5 L2 4 2
5. a) State and prove Cauchy's integral theorem. 6 L2 4 1
- b) Using Residue theorem, evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$, where C is the circle $|z| = \frac{5}{2}$. 5 L2 4 2
- c) Show that the function $f(z) = e^z$ is analytic everywhere. 5 L1 3 2
6. a) Obtain the Laurent's series expansion of $f(z) = \frac{z+3}{z(z^2-z-2)}$ in the region $|z| < 1$. 6 L2 4 2
- b) Define a harmonic function. Prove that the real and imaginary parts of an analytic function are harmonic. 5 L1 3 1
- c) Determine the analytic function $f(z)$ whose imaginary part is $v = e^x \sin y$. 5 L1 3 2

Unit – III

7. a) Find the Fourier series containing the first harmonics from the following table.

x	0	1	2	3	4	5
y	4	8	15	7	6	2

- b) Obtain the Fourier series of $f(x) = |x|$ in $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$. 6 L2 5 2
- c) Find the half range Fourier cosine series of $f(x) = x^2$ in $0 < x < \pi$. 5 L1 5 2
8. a) Using Unilateral Z-transforms, solve $u_{n+2} - 5u_{n+1} + 6u_n = (1)^n$ with $u_0 = 0, u_1 = 1$. 6 L2 2
- b) Find the bilateral Z-transform and region of convergence of $u_n = \begin{cases} 4^n, & \text{if } n < 0 \\ 2^n, & \text{if } n \geq 0 \end{cases}$. 5 L2 2
- c) State and prove Convolution theorem for bilateral Z-transforms. 5 L1 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
